

Idea Summary Report

Team 18

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1 Introduction

This document is the end-term evaluation report presented by **Team 18**. It includes the final updates, experimentation details, strategy overview, performance metrics, conclusions, and results of our high-frequency trading strategy project

2 Progress Update

Since the mid-semester stage, our work shifted from high-frequency PPO experiments toward more stable and risk-aware trading logic. Initial extensions such as Triple Barrier PPO and Expert Action Spoofing showed strong performance in frictionless settings but collapsed once realistic transaction costs were applied. This revealed a fundamental limitation: Reinforcement Learning agents were correctly predicting direction but operating at a frequency where average gains were smaller than execution costs, making the approach structurally unprofitable.

Recognizing this, we redirected efforts toward improving signal quality and risk control rather than forcing high-frequency trading. A multi-timescale regime-filtering framework was developed using price, volume, and band-based features. This system reliably identified trending vs. ranging markets and greatly reduced unnecessary exposure, achieving extremely low drawdowns even on low-volatility assets.

Building on these insights, the final strategy abandoned high-frequency behaviour entirely. Instead, it uses a disciplined, data driven process: a 3-hour statistical study phase to estimate trend and volatility, conservative percentile based thresholds for entries, and a strict one trade per day limit with mean based exits. This design preserved capital far better than earlier RL methods while still capturing meaningful intraday structure.

Overall, the project evolved from unstable high frequency RL to a robust, interpretable, and risk controlled rule-based system. The primary remaining challenge is refining dynamic stop-loss logic to prevent occasional adverse moves from offsetting accumulated gains.

3 Experimentation

This is the third section. Here, you can present the results of your work. You may want to include figures, tables, or charts to display your findings. Following the presentation of results, you can discuss their implications and what they mean in the context of your project.

3.1 Feature Engineering and Selection

To improve the signal quality and provide the agent with a more decisive view of the market, a detailed feature engineering process was undertaken.

3.1.1 Resampling

Tick by tick financial data is highly volatile and computationally intensive. To create a smoother and more meaningful

representation, we aggregated the data into **per minute intervals**. The resampling methods were tailored to each feature category:

- **Price-based Features:** Resampled into OHLC (Open, High, Low, Close) format for each minute.
- **Price Behavior (PB) Features:** The **mean** value within each one-minute interval was used.
- **Band Behavior (BB) Features:** The **last** available value in the interval was retained.
- **Volume based Features:** The **sum** of all tick level volumes within the minute was taken to represent total activity.
- **Other Derived Features:** Resampled using the **mean** value.

3.1.2 Normalization

To ensure all features were on a consistent scale, a **Standard-Scaler** was trained on the first 10-20 days of data. The resulting scaling parameters (mean and standard deviation) were then applied to the entire dataset, making the features comparable across different time periods and enhancing model stability.

3.1.3 Feature Selection Process

Feature selection was performed in multiple stages to retain only the most predictive and informative features.

Time Lagged and Cross Feature Analysis Very short time lagged features (below T4) were discarded as they primarily captured noise. We then generated derived features using time lagged differences within the same feature family (e.g., $PB1_T9 - PB1_T7$) to model the evolution between fast and slow moving signals. The discriminative power of each feature was evaluated by observing its ability to separate optimal long and short trading signals. Features that showed clear thresholds for separating bullish and bearish clusters were retained.

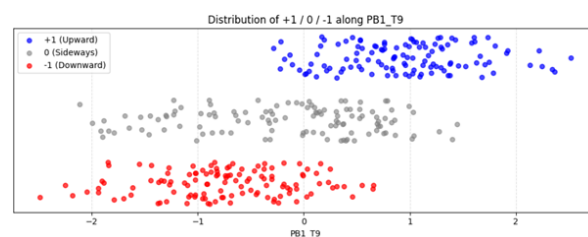


Figure 1: Feature based separation of long and short signals.

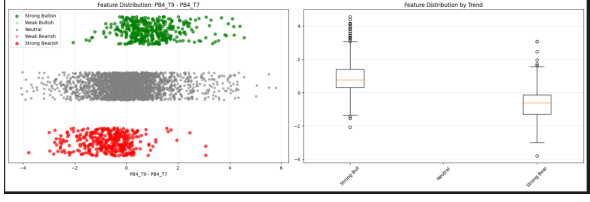


Figure 2: Example of a feature with strong discriminative power.

Volatility and Volume-Based Evaluation For volatility and volume features, we calculated feature wise accuracy scores based on how well they identified profitable trades within specific market regimes. The top performing features from all categories were selected for the final feature set.

Intra-Feature Transformations To capture momentum and reversal dynamics, we also generated intra feature time-lagged differences (e.g., $PB1_T7 - PB1_T7.\text{shift}(-k)$). This helped identify distinct regions of trend-following and mean-reversion behavior.

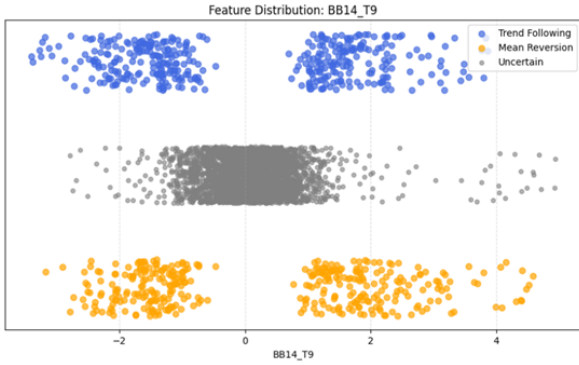


Figure 3: Identifying trend-following and mean-reversion regions.

By analyzing feature clusters, we selected the top N features that most clearly differentiated between these market behaviors.

3.2 Triple Barrier PPO

The Triple Barrier PPO experiment was designed to impose strict risk discipline on the Reinforcement Learning agent and mitigate the risks of overtrading in high-frequency environments. To achieve this, we implemented a "Triple Barrier" execution logic on the EBX asset, wrapping the agent's entry signals with rigid constraints: a fixed Take Profit level to secure momentum gains, a tight Stop Loss to cap downside volatility, and a Time Limit to force-close all positions before the end of the trading day. This hybrid structure allowed the PPO agent to focus exclusively on identifying high-probability entry points while the algorithmic wrappers managed the exits, effectively combining predictive learning with risk-aware execution. In a frictionless environment (zero transaction costs), the strategy produced exceptional

results: a Total PnL of +103.62%, a highly stable Max Draw-down of only 2.66%, and a Win Rate of 89.71%, confirming that the agent had successfully learned the EBX microstructure and could predict directional flows with near-perfect accuracy.

However, the introduction of realistic transaction costs exposed the limitations of high-frequency trading. With live-trade conditions, performance inverted, resulting in a Net PnL of -106.45% and a Win Rate collapse to 24.51%. This "Alpha-Cost Paradox" demonstrated that while the model accurately predicted the direction of price movements, the average gain per trade (0.04%) was frequently smaller than execution costs (0.02–0.05%), turning correct predictions into unprofitable micro-wins. Applying the same strategy to the lower-volatility EBY asset yielded minimal results, as the Triple Barrier parameters were rarely met, leading to few high-conviction entries. This confirmed that such high-frequency strategies are highly sensitive to asset-specific volatility and that their success relies on sufficient price movement to overcome transaction costs.

This experiment adhered to midterm trading constraints, with single-unit positions and no simultaneous holdings beyond -1 or +1. Overall, the Triple Barrier PPO strategy represents a methodically rigorous approach that balances predictive learning with disciplined execution, but also underscores the structural risks of friction and low-volatility environments in high-frequency trading.

3.3 Expert Action Spoofing (PPO + SDP)

The Expert Action Spoofing experiment was designed to address the "Policy Collapse" and inactivity observed in standard Reinforcement Learning. Recognizing that a vanilla PPO agent struggled with sparse and noisy reward signals, we developed a hybrid "Teacher-Student" architecture. In this framework, a Stochastic Dynamic Programming (SDP) solver acted as the Expert Teacher, computing the mathematically optimal sequence of trades the "Golden Path" based on either perfect foresight or high confidence Fractal Brownian Bridge forecasts. During training, the PPO agent's exploratory actions were overridden with the Expert's optimal moves while still observing the state and associated reward, effectively turning the problem into Behavioral Cloning or Demonstration Learning. This Action Spoofing approach forced the agent to internalize the Expert's decision making patterns, capture directional momentum, and learn to trade actively, without starting from random trial and error. During the trading phase, the Agent was encouraged to rely on its learned policy to approximate optimal behavior independently.

In frictionless simulations on the high volatility EBX asset, this method proved highly effective as the agent achieved a Total PnL of +29.46% with a Win Rate of 56.37%, significantly outperforming the Buy & Hold benchmark of 6.08%. The results validated that the neural network had partially internalized the Expert's logic and could capitalize on market volatility. However, the introduction of realistic transaction costs revealed a critical limitation. The Expert optimized

for total volatility capture, taking every available scalp, and the agent learned to mimic this hyperactive behavior. Under live fee conditions, the strategy became a "Churn Machine," generating excessive trade volume with average profits per trade smaller than execution costs, resulting in a Net PnL of -26.59% and a drop in Win Rate to 42.16%. On the lower volatility EBY asset, the strategy struggled even without costs (-5.19%), confirming that the model relies heavily on high volatility regimes for learnable signal generation.

Ultimately, the experiment was a predictive success but an execution failure. It demonstrated that a neural network can learn complex market dynamics from an optimal solver, yet ignoring transaction costs in the teaching phase leads to structurally unprofitable behavior. This experiment also adhered to midterm constraints, trading a single unit and never holding more than one position simultaneously.

3.4 Regime Filtering Strategy

The strategy integrates price, volume, and band indicators across multiple time horizons, allowing for robust detection of market regimes and informed trade decisions. Features are organized into long-term, medium-term, and short-term layers to capture structural, swing, and timing signals in the market.

For each horizon, aggregated feature scores such as $PriceScore_T$, $VolumeScore_T$, and $BandScore_T$ are computed, producing a smooth representation of market behaviour across timescales.

The time horizons are grouped into three conceptual layers:

- **Trend Layer:** captures long-term structural direction.
- **Swing Layer:** captures medium-term fluctuations.
- **Timing Layer:** captures short-term sentiment.

Within each layer, scores are averaged to summarise behaviour at that scale.

Composite indicators are constructed for each layer: Trend Composite (TC), Swing Composite (SC), and Timing Composite (TiC).

Each composite blends price, volume, and band-based scores:

$$TC = w_1 \cdot Price_Trend + w_2 \cdot Band_Trend + w_3 \cdot Volume_Trend$$

$$SC = w_4 \cdot Price_Swing + w_5 \cdot Band_Swing + w_6 \cdot Volume_Swing$$

$$TiC = w_7 \cdot Price_Timing + w_8 \cdot Band_Timing + w_9 \cdot Volume_Time$$

Directional alignment across TC, SC, and TiC is evaluated as:

$$DA = sign(TC) + sign(SC) + sign(TiC)$$

Strong positive DA indicates consistent upward movement, strong negative DA indicates downward momentum, and

mixed readings imply uncertainty.

Regime detection classifies the market as **Ranging** or **Trending** based on composite strengths and alignment. Confidence scores (0–1) are assigned to quantify reliability. Position sizing depends on regime confidence, trend strength, volume confirmation, and price-volume alignment, ensuring larger positions in stable trends and smaller positions in weak or noisy conditions.

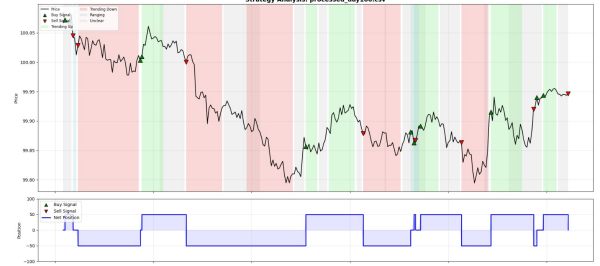


Figure 4: Image depicting Regime Filtering .

Entry occurs when three conditions align:

1. Regime-based confirmation (Trending regime with stable direction and sufficient confidence)
2. Price-volume confirmation (volume supports trend direction)
3. Position sizing agreement (scaled according to trend strength and confidence)

Exit mechanisms include:

- ATR-based stop loss: $SL = Price \pm ATR_mult \cdot ATR \cdot (1 + RegimeStrength)$
- Regime-based exit: triggered when market shifts to Ranging or confidence drops
- Trend reversal / volume divergence exit: triggered when regime direction flips or volume diverges from TC

This multi-layer framework integrates price, volume, and band signals, reduces false signals, and enhances risk-adjusted returns across different market conditions. When evaluated on Asset EBY under realistic transaction cost assumptions, the strategy produced an exceptionally low Max Drawdown of 0.18%, demonstrating its effectiveness as a defensive filter. It correctly identified most periods as non-tradable, avoiding unnecessary market exposure. Although the overall return was -3.07% due to the asset's inherently low volatility, the experiment validated the strength, clarity, and robustness of the regime-detection methodology. The results confirm that the approach is methodologically sound and highly effective at capital preservation, with profitability dependent on the volatility characteristics of the traded instrument.

4 Strategy Overview

This is a conservative, regime based quantitative strategy designed to execute a few high conviction trade per day to prioritize capital preservation and minimize drawdown. It utilizes an initial 90 minute "study phase" to establish statistical baselines calculating mean, standard deviation, and trend direction via linear regression while simultaneously applying a strict volatility filter to reject hazardous, high-range market conditions. Based on the identified regime (Uptrend, Downtrend, or Flat), the algorithm dynamically sets entry thresholds at specific statistical percentiles (e.g., buying deep pullbacks in an uptrend) and targets mean reversion based exits, ensuring disciplined execution with a mandatory hard-stop liquidation at the end of the session (06:10:00)

4.1 Core Philosophy and Capital Preservation

The fundamental premise of this strategy is "Quality over Quantity." Analysis of previous backtests revealed a recurring pattern where subsequent trades on the same day frequently resulted in significant losses, eroding the alpha generated by the initial setup. Consequently, this algorithm adopts a strict Single-Shot constraint: it executes a maximum of one trade per day. This limitation forces the model to be highly selective, prioritizing high-conviction setups derived from statistical anomalies while strictly limiting exposure to prolonged market noise.

4.2 The Market Acquisition Phase (Study Period)

At the beginning of every trading session, the strategy enters a passive 90-minute Study Phase (5400 seconds). During this window, the algorithm places no orders. Instead, it aggregates real-time tick data to build a statistical profile of the specific day's market behavior. By waiting 1.5 hours, the model avoids the chaotic volatility often seen at the market open and ensures that its entry parameters are calibrated to the current day's specific volatility and trend profile, rather than relying on historical averages that may be obsolete.

4.3 Regime Classification and Volatility Filtering

Once the study phase ends, the strategy performs a two-step analysis to determine the market regime. First, it applies a **Volatility Gatekeeper**. The model computes the absolute price range,

$$\text{Range} = \text{High} - \text{Low},$$

and if this value exceeds 0.60, the market is classified as **Too Volatile / Unpredictable**, disabling trading for the rest of the day. This serves as a circuit breaker against high-variance conditions that typically increase stop-out probability.

If volatility is within acceptable bounds, the model proceeds with **Trend Detection** by fitting a linear regression to the price series. From this regression, it extracts the **Slope** (trend

direction), the coefficient of determination R^2 (trend quality), and a simple **Momentum** metric:

$$\text{Momentum} = \text{End Price} - \text{Start Price}.$$

Using these metrics, the strategy classifies the prevailing market regime as:

- **Uptrend:** Positive slope, high R^2 , and positive momentum.
- **Downtrend:** Negative slope, high R^2 , and negative momentum.
- **Flat:** Any case that does not meet the strict trend conditions.

4.4 Dynamic Statistical Thresholding

Unlike strategies that use fixed price levels, this model generates dynamic entry and exit barriers based on the statistical distribution (Percentiles and Standard Deviation) of the study data:

- **In an Uptrend:** The strategy adopts a "Buy the Dip" logic. It sets a conservative Buy limit at the 15th percentile of the study prices (a deep pullback) and sets the Target at the Mean + 1 Standard Deviation. It refuses to open Short positions.
- **In a Downtrend:** The strategy adopts a "Sell the Rally" logic. It sets a Short Entry at the 85th percentile (a high extension) and covers at the Mean - 1 Standard Deviation. It refuses to open Long positions.
- **In a Flat Market:** It plays pure Mean Reversion with tighter bands, buying at the 12th percentile and selling at the 88th percentile.

4.5 Execution and Risk Management

Once the thresholds are set, the strategy switches to Active Mode. It monitors the live feed for the price to cross the entry threshold. If triggered, it enters a position size of 100 units.

- **Profit Taking:** The strategy utilizes a "Reversion to Mean" exit logic. If the price returns to the calculated statistical mean (or crosses the opposing threshold), the position is liquidated immediately to lock in profits.
- **Hard Constraints:** The strategy enforces a hard time-based exit at **06:10:00** to avoid the predefined penalty due to intraday trade. Regardless of Profit /Loss or technical indicators, all open positions are forcefully squared off at this time to prevent end-of-session liquidity risks. Once a trade is closed (whether by Target or Time), the strategy locks itself out until the next day.

4.6 Conservative Statistical Thresholds

4.6.1 Rationale: The Pivot from Prediction to Management

Our exhaustive testing of Reinforcement Learning and Causal Regime Detection led to a singular, unavoidable conclusion regarding the provided market environment: **High-frequency interaction is statistically ruinous.**

While our previous models achieved high directional accuracy (85%+ Win Rate), they failed due to churn. Taking 50 trades a day multiplies the transaction cost hurdle by 50, requiring an impossible level of volatility to break even. Consequently, we pivoted our entire philosophy from **Maximizing Alpha Capture** (getting every move) to **Minimizing Friction Exposure** (taking only the best move). We developed the **Conservative Single-Trade Strategy** based on three core intuitions:

1. **The "One Shot" Rule:** By limiting the agent to a maximum of one trade per day, we mathematically cap our transaction cost exposure to the absolute minimum. This forces the algorithm to be "hyper-selective," trading only when statistical probability is at its peak
2. **Context over History:** Rather than training on months of historical data (which may be stale), this strategy uses a 90-minute "Study Phase" at the start of every day. It calculates the current day's mean, standard deviation, and trend profile before placing a single order. It adapts to the market as it is today.
3. **Statistical Reversion:** Instead of predicting where price will go, we identify where price is relative to its daily distribution. We buy only when the price is in the 15th Percentile (Deep Dip) of an Uptrend, and sell only in the 85th Percentile (High Extension) of a Downtrend.

4.7 Comprehensive Summary

The trading framework integrates statistical preprocessing, regime identification, and rule-based execution into a unified intraday decision system. The process begins with extracting the first 1.5 hours of tick-level data to compute compact statistical descriptors mean, standard deviation, price range, regression slope, R^2 , momentum, and key percentile bands. These descriptors summarize early-session market structure and serve as the basis for adaptive threshold construction.

Using these statistics, the system classifies the session into one of three regimes: uptrend, downtrend, or flat. Each regime defines a dedicated entry model pullback buys, rally-based shorts, or mean-reversion bands. Thresholds are reinforced with spacing constraints, volatility filters, and stability checks, and days exhibiting excessive initial volatility are excluded to avoid unfavourable conditions.

After the study phase, the strategy transitions to an event-driven execution loop. Every incoming tick is evaluated through the decision function, which maps live price to a target position. The backtesting layer converts position

deltas into trades, manages order execution, and updates PnL via a dedicated callback. A strict **one-trade-per-day** constraint ensures capital preservation and prevents overreaction to intraday noise.

Risk is controlled through a hard liquidation at 06:10:00, ensuring no positions carry into the next session. Additional safeguards state resets, deterministic order handling, and regime-specific exit logic maintain consistency and prevent unintended exposure. Together, these components form a disciplined, interpretable, and conservative intraday system that blends statistical rigor with controlled execution.

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